

Análisis Comparativo de Gini y Entropía

Este notebook compara el Índice de Gini y la Entropía para dos variables categóricas relevantes en dos periodos de tiempo.

1. Importar librerías necesarias

```
# Importar librerías
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, to_date
from pyspark.sql.types import NumericType
import os
import numpy as np
import matplotlib.pyplot as plt
```

2. Inicializar SparkSession

```
# Inicializar SparkSession
spark =
SparkSession.builder.appName('GiniEntropiaComparativo').getOrCreate()
```

3. Cargar los datos desde la carpeta merge_pyspark

```
# Cargar todos los archivos Parquet de la carpeta merge_pyspark
parquet_folder = '../merge_pyspark'
parquet_files = [os.path.join(parquet_folder, f) for f in
os.listdir(parquet_folder) if f.endswith('.parquet')]
df = spark.read.parquet(*parquet_files)
df.show(5)
df.printSchema()
```

```
+-----+-----+-----+-----+-----+
|      customer_id|article_id|      Fecha|      price|
sales_channel_id|product_code|      prod_name|product_type_no|
product_type_name|product_group_name|graphical_appearance_no|
graphical_appearance_name|colour_group_code|colour_group_name|
perceived_colour_value_id|perceived_colour_value_name|
perceived colour master id|perceived colour master name|department no|
```

department_name	index_code		index_name	index_group_no		
index_group_name	section_no		section_name	garment_group_no		
garment_group_name		detail_desc	FN	Active		
club_member_status	fashion_news_frequency	age			postal_code	
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+-----+						
+-----+						
00030bc026428965d...	673677015	2019-11-28	0.0254067796610169			
2	673677	Henry polo (1)	252		Sweater	
Garment Upper body		1010016			Solid	
73	Dark Blue		4			
Dark		2		Blue		
1616	Knitwear	A	Ladieswear		1	
Ladieswear	11	Womens Tailoring		1003		
Knitwear Jumper in a soft,...	NULL	NULL		ACTIVE		
NONE	57 2c29ae653a9282cce...					
00030bc026428965d...	673677010	2019-11-28	0.0254067796610169			
2	673677	Henry polo (1)	252		Sweater	
Garment Upper body		1010016			Solid	
43	Dark Red		4			
Dark		18		Red		
1616	Knitwear	A	Ladieswear		1	
Ladieswear	11	Womens Tailoring		1003		
Knitwear Jumper in a soft,...	NULL	NULL		ACTIVE		
NONE	57 2c29ae653a9282cce...					
00083cda041544b2f...	683001006	2019-02-10	0.0169322033898305			
2	683001	Pattern 7p Socks	302		Socks	
Socks & Tights		1010001		All over pattern		
10	White		3			
Light		9		White		
3611 Shopbasket Socks		B Lingeries/Tights			1	
Ladieswear	62 Womens Nightwear,...			1021	Socks	
and Tights Fine-knit socks i...	NULL	NULL		ACTIVE		
NONE	29 24e3594738f327e8a...					
00083cda041544b2f...	568571002	2018-10-24	0.0135423728813559			
2	568571 Spartacus cheeky ...		59	Swimwear bottom		
Swimwear		1010016		Solid		
9	Black		4			
Dark		5		Black		
4242	Swimwear	B Lingeries/Tights			1	
Ladieswear	60 Womens Swimwear, ...			1018		
Swimwear Fully lined, text...	NULL	NULL		ACTIVE		

```

NONE| 29|24e3594738f327e8a...|
|00083cda041544b2f...| 723469007|2019-02-10|0.0254067796610169|
2| 723469|Kelly Push (Melbo...| 306| Bra|
Underwear| 1010016| Solid|
8| Dark Grey| 4|
Dark| 12| Grey|
1334| Casual Lingerie| B|Lingeries/Tights| 1|
Ladieswear| 61| Womens Lingerie| 1017| Under-,
Nightwear|Push-up bras in s...|NULL| NULL| ACTIVE|
NONE| 29|24e3594738f327e8a...|

```

```

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+-----+-----+-----+-----+
+-----+-----+-----+-----+

```

only showing top 5 rows

root

```

|-- customer_id: string (nullable = true)
|-- article_id: long (nullable = true)
|-- Fecha: date (nullable = true)
|-- price: double (nullable = true)
|-- sales_channel_id: long (nullable = true)
|-- product_code: integer (nullable = true)
|-- prod_name: string (nullable = true)
|-- product_type_no: integer (nullable = true)
|-- product_type_name: string (nullable = true)
|-- product_group_name: string (nullable = true)
|-- graphical_appearance_no: integer (nullable = true)
|-- graphical_appearance_name: string (nullable = true)
|-- colour_group_code: integer (nullable = true)
|-- colour_group_name: string (nullable = true)
|-- perceived_colour_value_id: integer (nullable = true)
|-- perceived_colour_value_name: string (nullable = true)
|-- perceived_colour_master_id: integer (nullable = true)
|-- perceived_colour_master_name: string (nullable = true)
|-- department_no: integer (nullable = true)
|-- department_name: string (nullable = true)
|-- index_code: string (nullable = true)
|-- index_name: string (nullable = true)
|-- index_group_no: integer (nullable = true)
|-- index_group_name: string (nullable = true)
|-- section_no: integer (nullable = true)
|-- section_name: string (nullable = true)

```

```

|-- garment_group_no: integer (nullable = true)
|-- garment_group_name: string (nullable = true)
|-- detail_desc: string (nullable = true)
|-- FN: double (nullable = true)
|-- Active: double (nullable = true)
|-- club_member_status: string (nullable = true)
|-- fashion_news_frequency: string (nullable = true)
|-- age: integer (nullable = true)
|-- postal_code: string (nullable = true)

```

4. Preprocesamiento y conversión de fechas

```

# Convertir la columna de fecha al formato adecuado y eliminar nulos
df = df.withColumn('fecha', to_date(col('fecha'), 'dd/MM/yyyy'))
df = df.dropna(subset=['fecha'])
df.select('fecha').show(5)

```

```

+-----+
|      fecha|
+-----+
|2019-11-28|
|2019-11-28|
|2019-02-10|
|2018-10-24|
|2019-02-10|
+-----+
only showing top 5 rows

```

5. División de los datos en dos conjuntos por rango de fechas

```

# Definir los rangos de fechas
from datetime import date
inicio_1 = date(2018, 9, 20)
fin_1 = date(2019, 12, 31)
inicio_2 = date(2020, 1, 1)
fin_2 = date(2020, 9, 22)

conjunto1 = df.filter((col('fecha') >= inicio_1) & (col('fecha') <= fin_1))
conjunto2 = df.filter((col('fecha') >= inicio_2) & (col('fecha') <= fin_2))
print('Conjunto 1:', conjunto1.count(), 'registros')
print('Conjunto 2:', conjunto2.count(), 'registros')

Conjunto 1: 20808192 registros
Conjunto 2: 10980132 registros

```

6. Selección de variables categóricas para el análisis

Se analizarán `product_group_name` y `department_name`.

```
# Variables categóricas a comparar
variables_categoricas = ['product_group_name', 'department_name']

def get_frequencies(df, colname):
    return df.groupby(colname).count().toPandas()

freqs1 = {var: get_frequencies(conjunto1, var) for var in
variables_categoricas}
freqs2 = {var: get_frequencies(conjunto2, var) for var in
variables_categoricas}

for var in variables_categoricas:
    print(f'Frecuencias {var} - Conjunto 1:')
    print(freqs1[var])
    print(f'Frecuencias {var} - Conjunto 2:')
    print(freqs2[var])
```

Frecuencias `product_group_name` - Conjunto 1:

	product_group_name	count
0	Garment Lower body	4599637
1	Garment and Shoe care	64
2	Garment Upper body	8610456
3	Swimwear	1484678
4	Bags	3162
5	Items	2149
6	Unknown	26017
7	Underwear/nightwear	473
8	Accessories	1092446
9	Furniture	456
10	Socks & Tights	481437
11	Cosmetic	1311
12	Underwear	1592224
13	Nightwear	241750
14	Shoes	504204
15	Garment Full body	2167674
16	Interior textile	54

Frecuencias `product_group_name` - Conjunto 2:

	product_group_name	count
0	Garment Lower body	2446417
1	Garment and Shoe care	215
2	Garment Upper body	3942299
3	Stationery	229
4	Swimwear	1094544
5	Bags	4151
6	Items	3278
7	Unknown	71023

```

8      Underwear/nightwear      86
9          Accessories      507147
10         Furniture      77
11      Socks & Tights      204275
12         Cosmetic      189
13         Underwear      973634
14         Nightwear      106430
15         Shoes      241317
16         Fun      5
17      Interior textile      20
18      Garment Full body      1384796
Frecuencias department_name - Conjunto 1:
      department_name      count
0      Jacket Casual      8730
1      Denim Trousers      446430
2      Dresses      460075
3      Basic 1      688477
4      Kids Girl Outdoor      6700
..      ...
238      Kids Girl License      17
239      Suit Extended inactive from s1      9
240      Kids Boy License      15
241      Baby Girl Local Relevance      9
242      Shirt Extended inactive from s1      1

```

[243 rows x 2 columns]

```

Frecuencias department_name - Conjunto 2:
      department_name      count
0      Jacket Casual      3186
1      Dresses      259111
2      Denim Trousers      229512
3      Basic 1      452122
4      Young Girl Knitwear      2103
..      ...
239      Promotion / Other / Offer      82
240      EQ Divided Blue      66
241      Sneakers small girl inactive from s2      2
242      Small Acc. Jewellery & Other      4
243      Kids Girl License      1

```

[244 rows x 2 columns]

7. Cálculo del Índice de Gini y Entropía para ambas variables

```

# Funciones para calcular Gini y Entropía
def gini_index(counts):
    probs = counts / np.sum(counts)
    return 1 - np.sum(probs ** 2)

```

```
def entropy_index(counts):
    probs = counts / np.sum(counts)
    return -np.sum(probs * np.log2(probs + 1e-9))

resultados = {}
for var in variables_categoricas:
    counts1 = freqs1[var]['count'].values
    counts2 = freqs2[var]['count'].values
    resultados[var] = {
        'gini1': gini_index(counts1),
        'gini2': gini_index(counts2),
        'entropia1': entropy_index(counts1),
        'entropia2': entropy_index(counts2)
    }
    print(f"Variable: {var}")
    print(f"  Gini Conjunto 1: {resultados[var]['gini1']:.4f}")
    print(f"  Gini Conjunto 2: {resultados[var]['gini2']:.4f}")
    print(f"  Entropía Conjunto 1: {resultados[var]
['entropia1']:.4f}")
    print(f"  Entropía Conjunto 2: {resultados[var]
['entropia2']:.4f}")

Variable: product_group_name
  Gini Conjunto 1: 0.7541
  Gini Conjunto 2: 0.7846
  Entropía Conjunto 1: 2.4742
  Entropía Conjunto 2: 2.5852
Variable: department_name
  Gini Conjunto 1: 0.9719
  Gini Conjunto 2: 0.9664
  Entropía Conjunto 1: 5.9817
  Entropía Conjunto 2: 5.6777
```

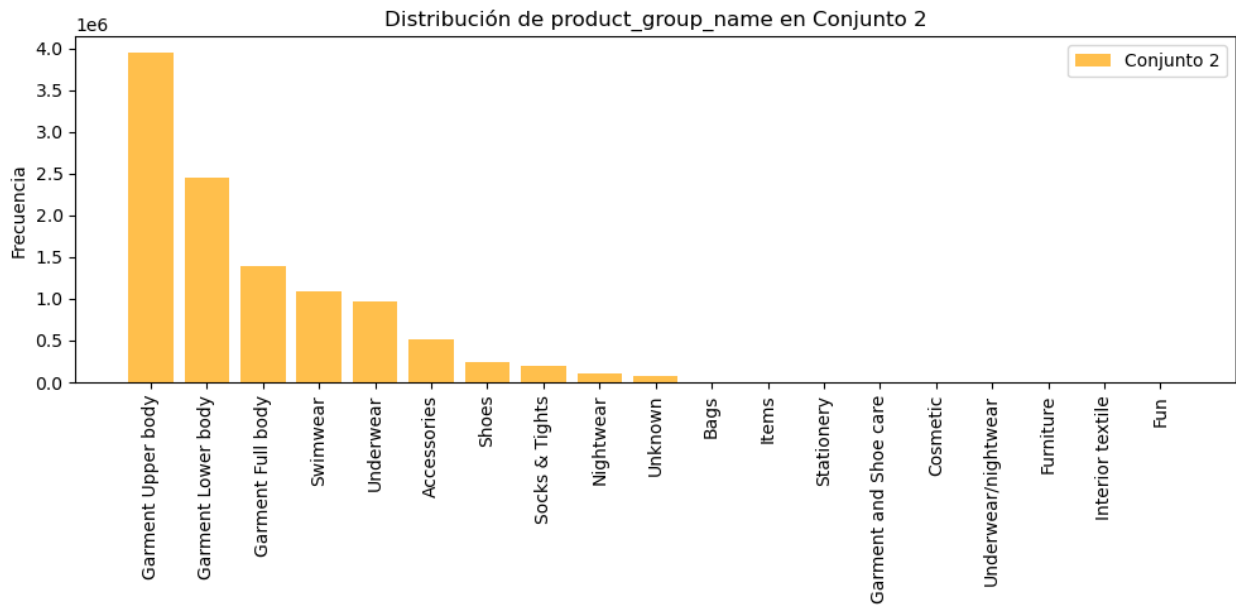
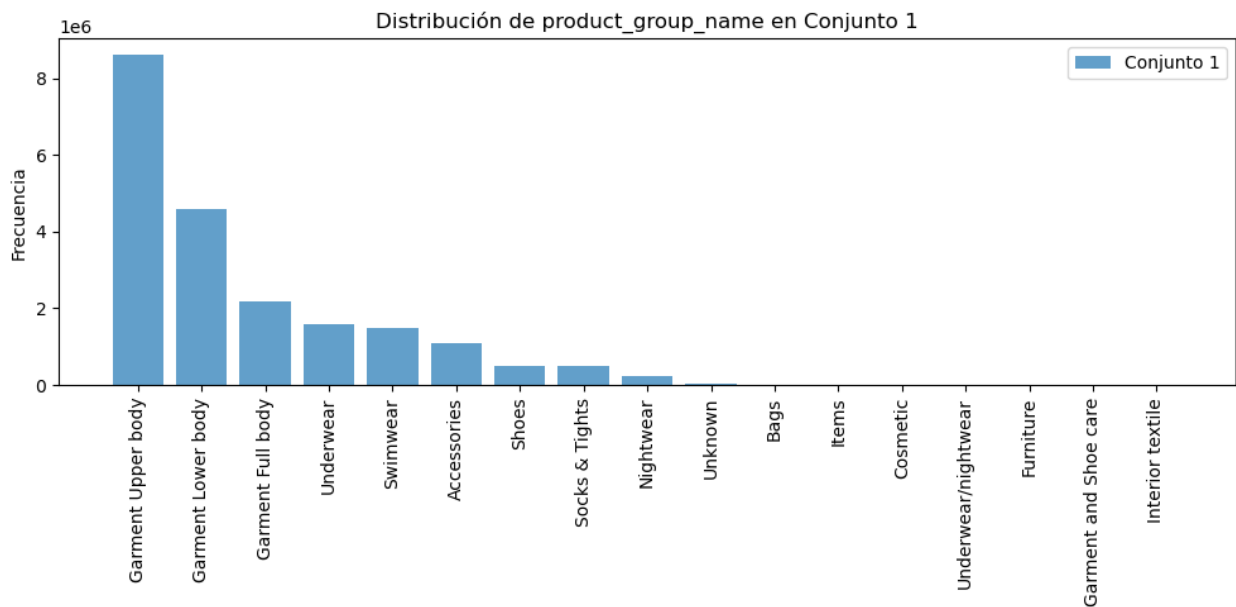
8. Visualización comparativa de la distribución de ambas variables

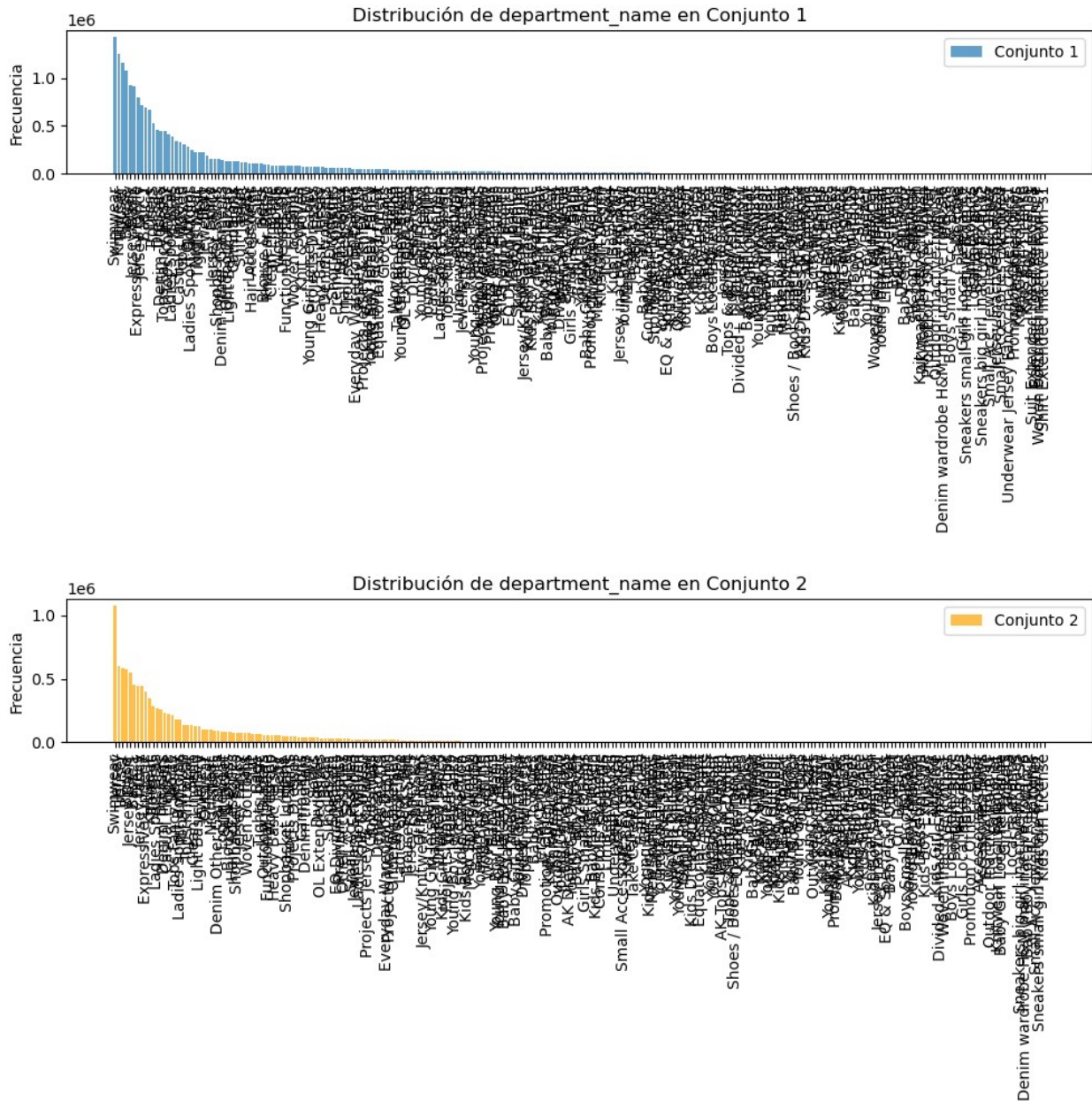
```
for var in variables_categoricas:
    plt.figure(figsize=(10,5))
    freq1_sorted = freqs1[var].sort_values('count', ascending=False)
    plt.bar(freq1_sorted[var], freq1_sorted['count'], alpha=0.7,
label=f'Conjunto 1')
    plt.xticks(rotation=90)
    plt.title(f'Distribución de {var} en Conjunto 1')
    plt.ylabel('Frecuencia')
    plt.legend()
    plt.tight_layout()
    plt.show()
```

```

plt.figure(figsize=(10,5))
freq2_sorted = freqs2[var].sort_values('count', ascending=False)
plt.bar(freq2_sorted[var], freq2_sorted['count'], alpha=0.7,
color='orange', label=f'Conjunto 2')
plt.xticks(rotation=90)
plt.title(f'Distribución de {var} en Conjunto 2')
plt.ylabel('Frecuencia')
plt.legend()
plt.tight_layout()
plt.show()

```





9. Interpretación comparativa de los resultados

- El **Índice de Gini** y la **Entropía** permiten comparar la diversidad y concentración de las variables categóricas seleccionadas en ambos periodos.
- Si una variable muestra mayor Gini y Entropía en un periodo, indica mayor diversidad y menor concentración en categorías específicas.